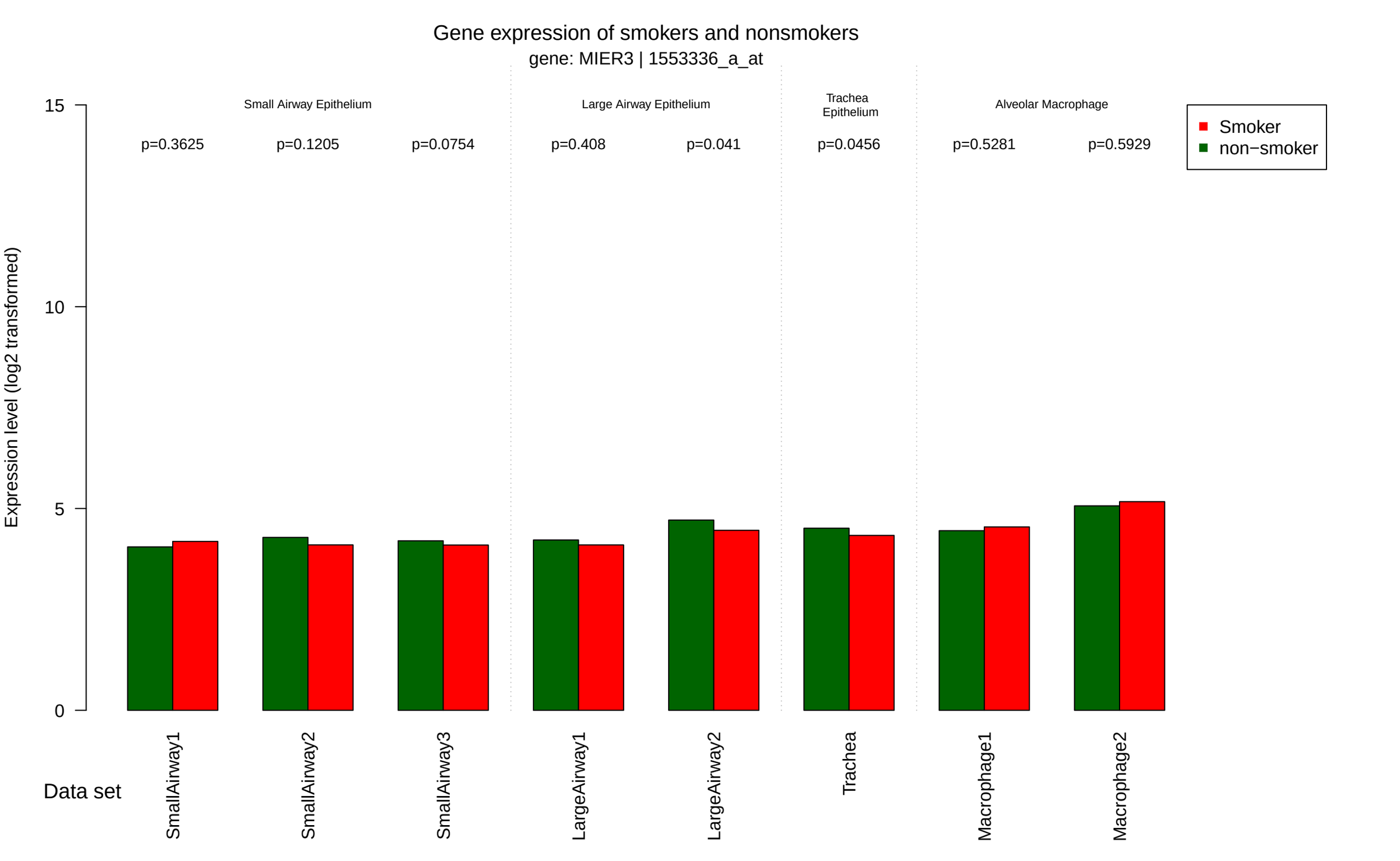
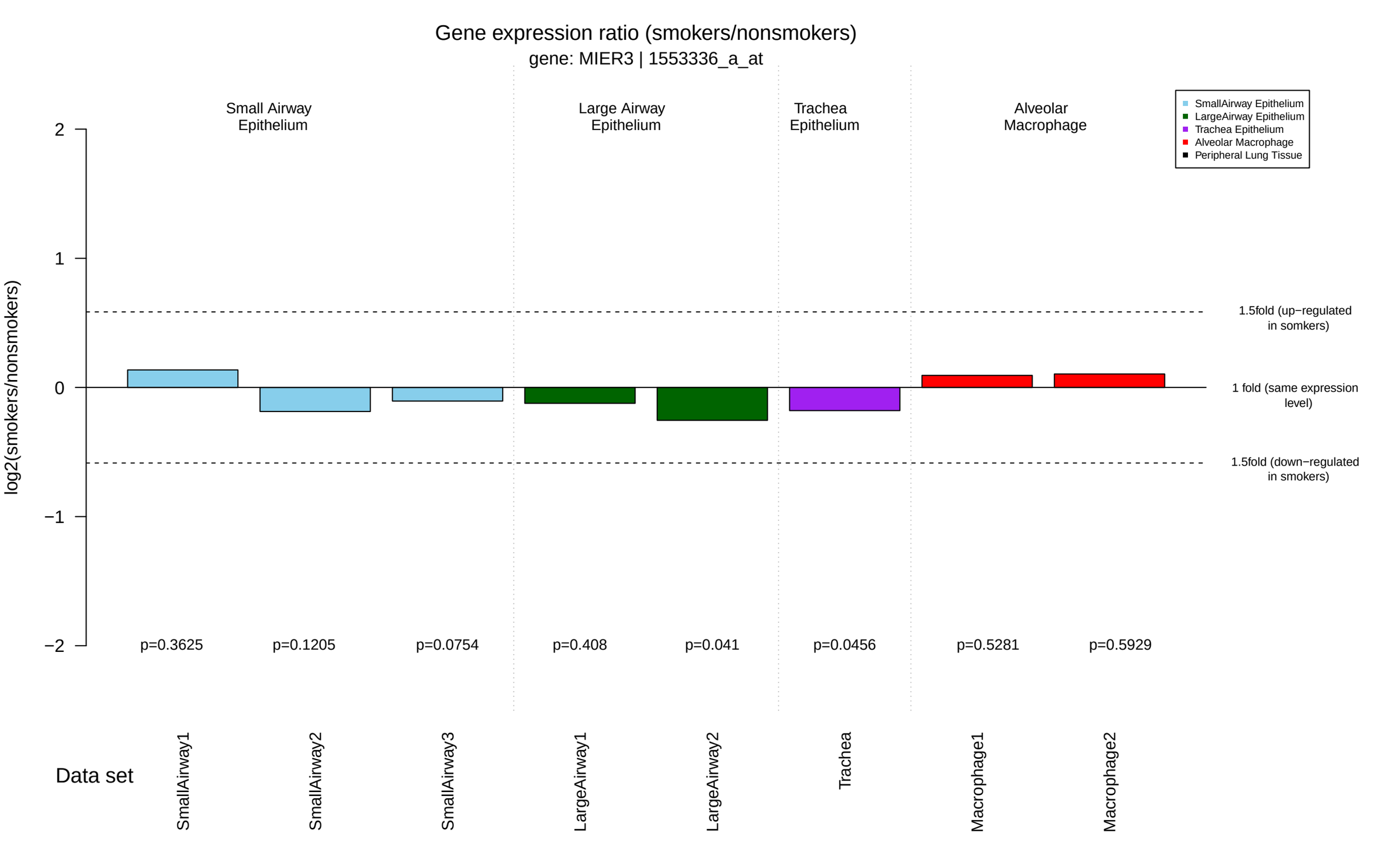
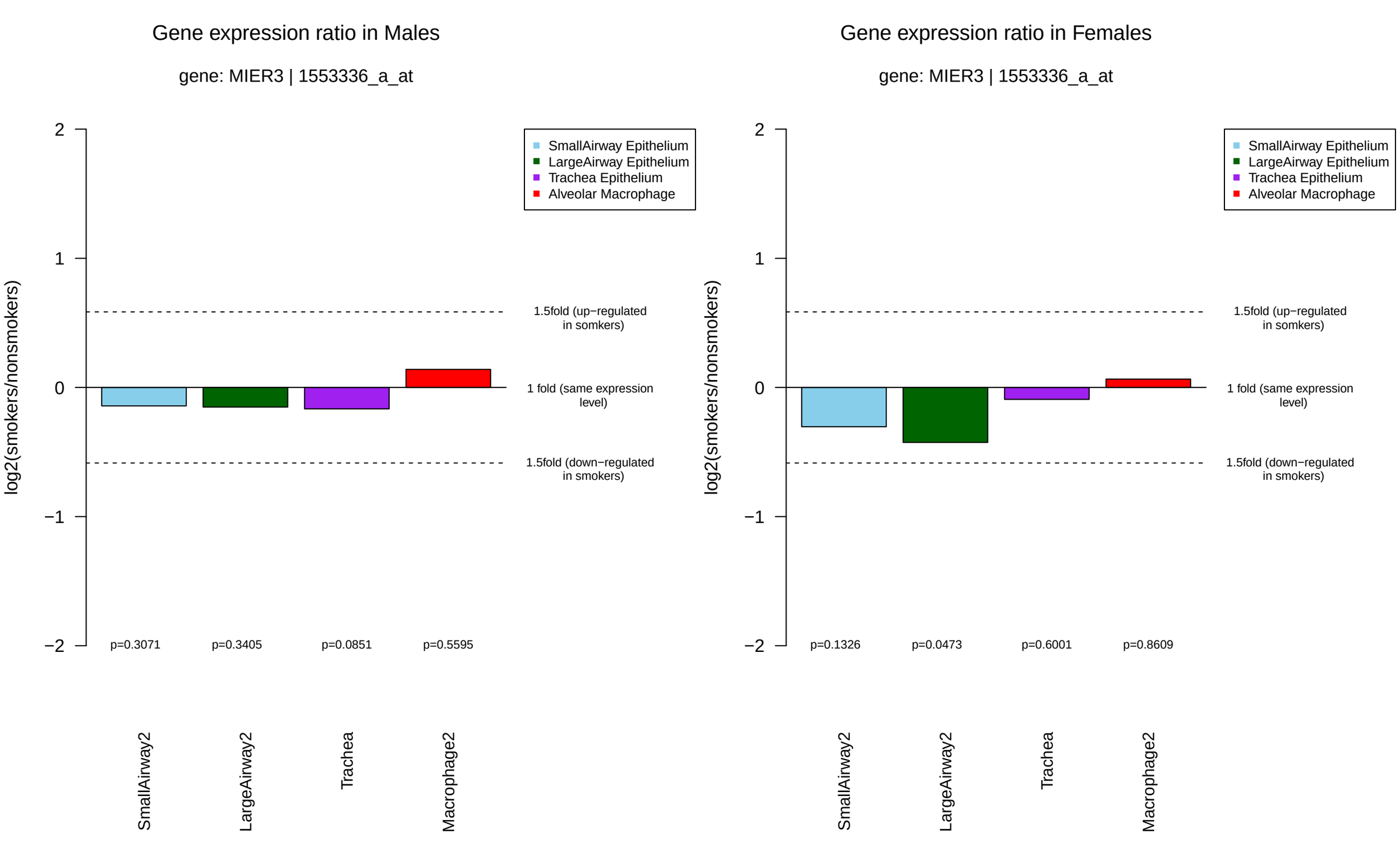
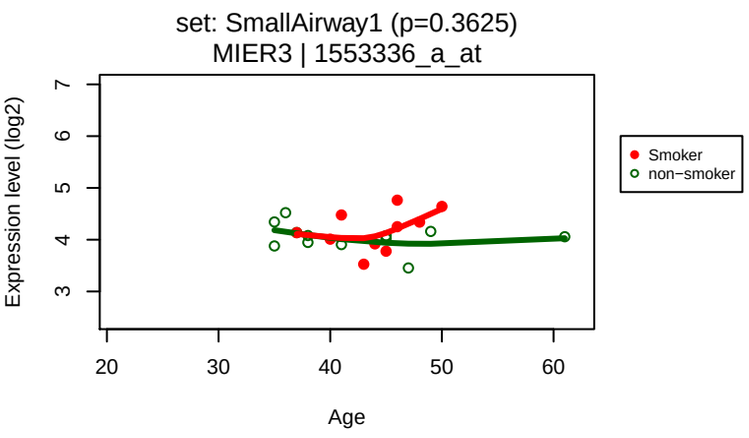
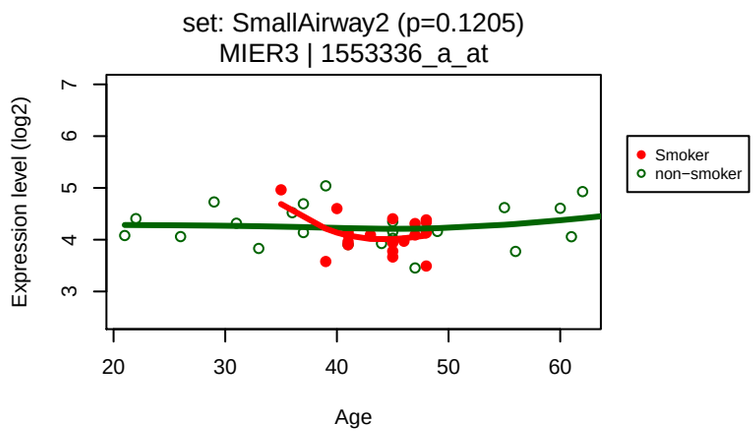
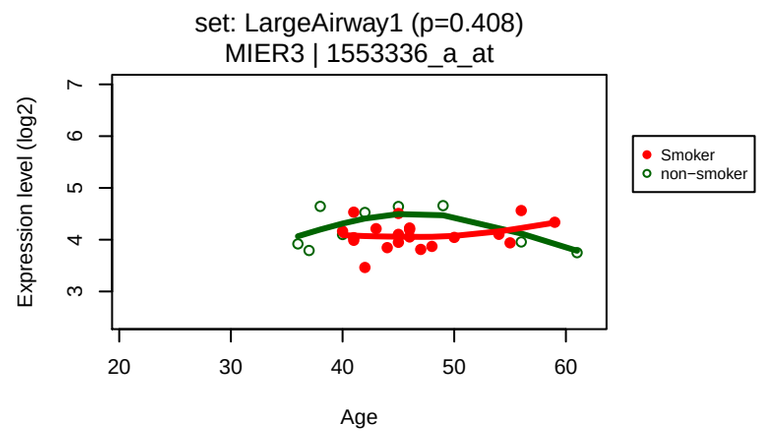

Gene expression of smokers and nonsmokers
gene: MIER3 | 1553336_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3625
p=0.1205
p=0.0754
p=0.408
p=0.041
p=0.0456
p=0.5281
p=0.5929
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

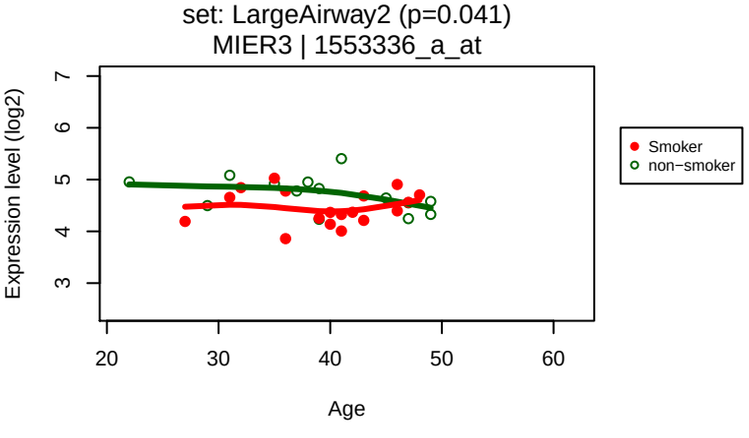

Gene expression ratio (smokers/nonsmokers)
gene: MIER3 | 1553336_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3625
p=0.1205
p=0.0754
p=0.408
p=0.041
p=0.0456
p=0.5281
p=0.5929
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

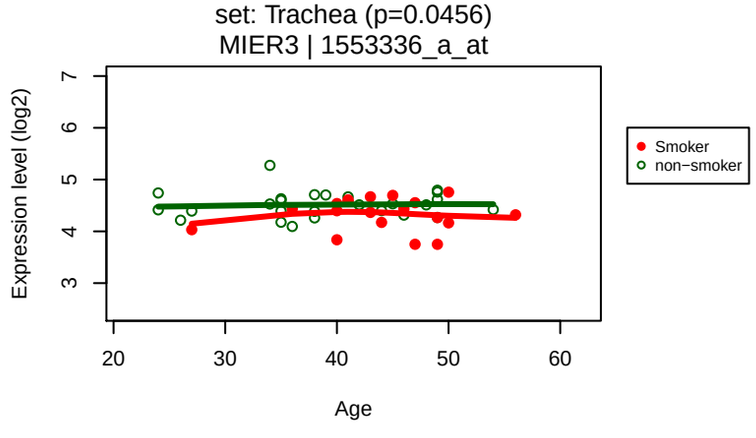

Gene expression ratio in Males
gene: MIER3 | 1553336_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3071
p=0.3405
p=0.0851
p=0.5595
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MIER3 | 1553336_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1326
p=0.0473
p=0.6001
p=0.8609
SmallAirway2
LargeAirway2
Trachea
Macrophage2

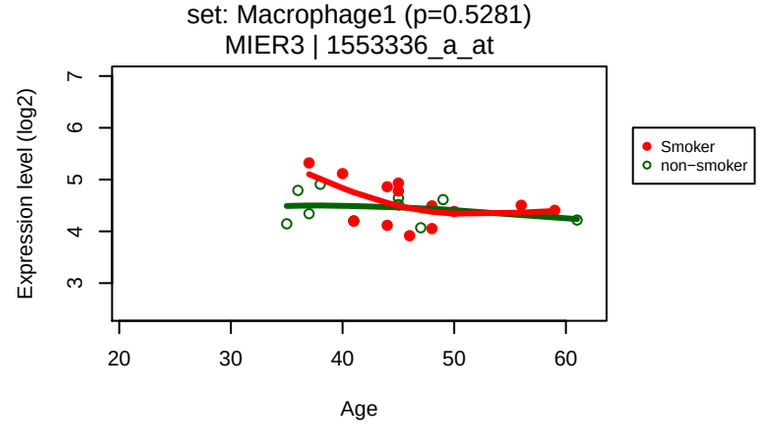

set: SmallAirway1 (p=0.3625)
MIER3 | 1553336_a_at
Expression level (log2)
Age
Smoker
non−smoker

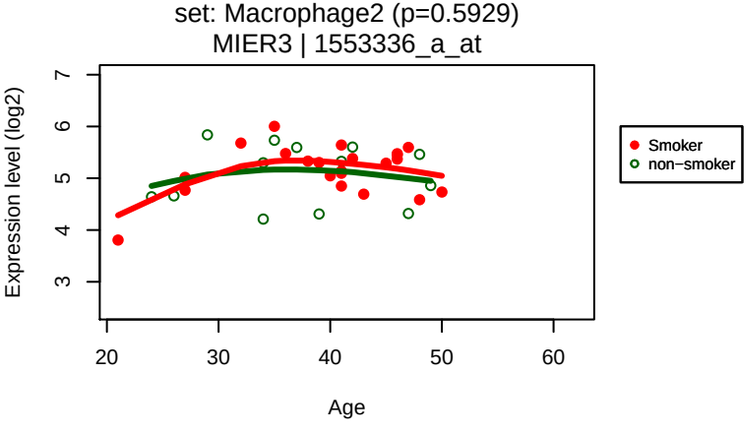

set: SmallAirway2 (p=0.1205)
MIER3 | 1553336_a_at
Expression level (log2)
Age
Smoker
non−smoker

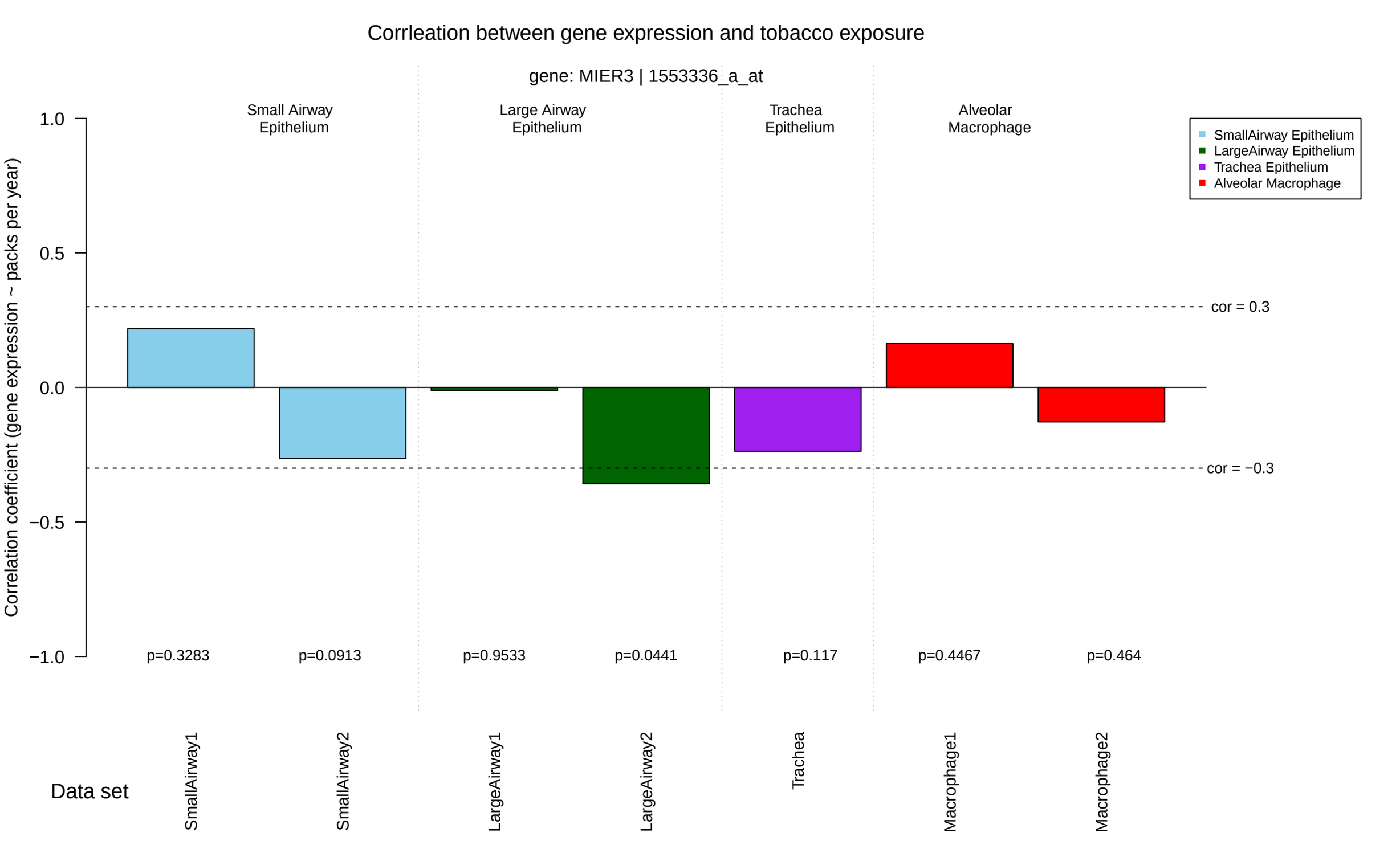

set: LargeAirway1 (p=0.408)
MIER3 | 1553336_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.041)
MIER3 | 1553336_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0456)
MIER3 | 1553336_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5281)
MIER3 | 1553336_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5929)
MIER3 | 1553336_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MIER3 | 1553336_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3283
p=0.0913
p=0.9533
p=0.0441
p=0.117
p=0.4467
p=0.464
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2